

tf.compat.v1.losses.mean_pairwise_square

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https://www.tensorflow.org/api_docs/python/tf/compat/v1/losses/mean_pairwise_squared_error

Adds a pairwise-errors-squared loss to the training procedure.

Function Signatures

```
tf.compat.v1.losses.mean_pairwise_squared_error( labels,  
predictions, weights=1.0, scope=None,  
loss_collection=ops.GraphKeys.LOSSES )
```

Code Examples

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    labels,  
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Documentation

Adds a pairwise-errors-squared loss to the training procedure.

Unlike `mean_squared_error`, which is a measure of the differences between corresponding elements of predictions and labels, `mean_pairwise_squared_error` is a measure of the differences between pairs of corresponding elements of predictions and labels.

For example, if `labels=[a, b, c]` and `predictions=[x, y, z]`, there are three pairs of differences are summed to compute the loss:

$$\text{loss} = [((a-b) - (x-y)).^2 + ((a-c) - (x-z)).^2 + ((b-c) - (y-z)).^2] / 3$$

Note that since the inputs are of shape `[batch_size, d0, ... dN]`, the

corresponding pairs are computed within each batch sample but not across samples within a batch. For example, if `predictions` represents a batch of 16 grayscale images of dimension `[batch_size, 100, 200]`, then the set of pairs is drawn from each image, but not across images.

`weights` acts as a coefficient for the loss. If a scalar is provided, then the loss is simply scaled by the given value. If `weights` is a tensor of size `[batch_size]`, then the total loss for each sample of the batch is rescaled by the corresponding element in the `weights` vector.

Returns

Raises

eager compatibility

The `loss_collection` argument is ignored when executing eagerly. Consider holding on to the return value or collecting losses via a `tf.keras.Model`.